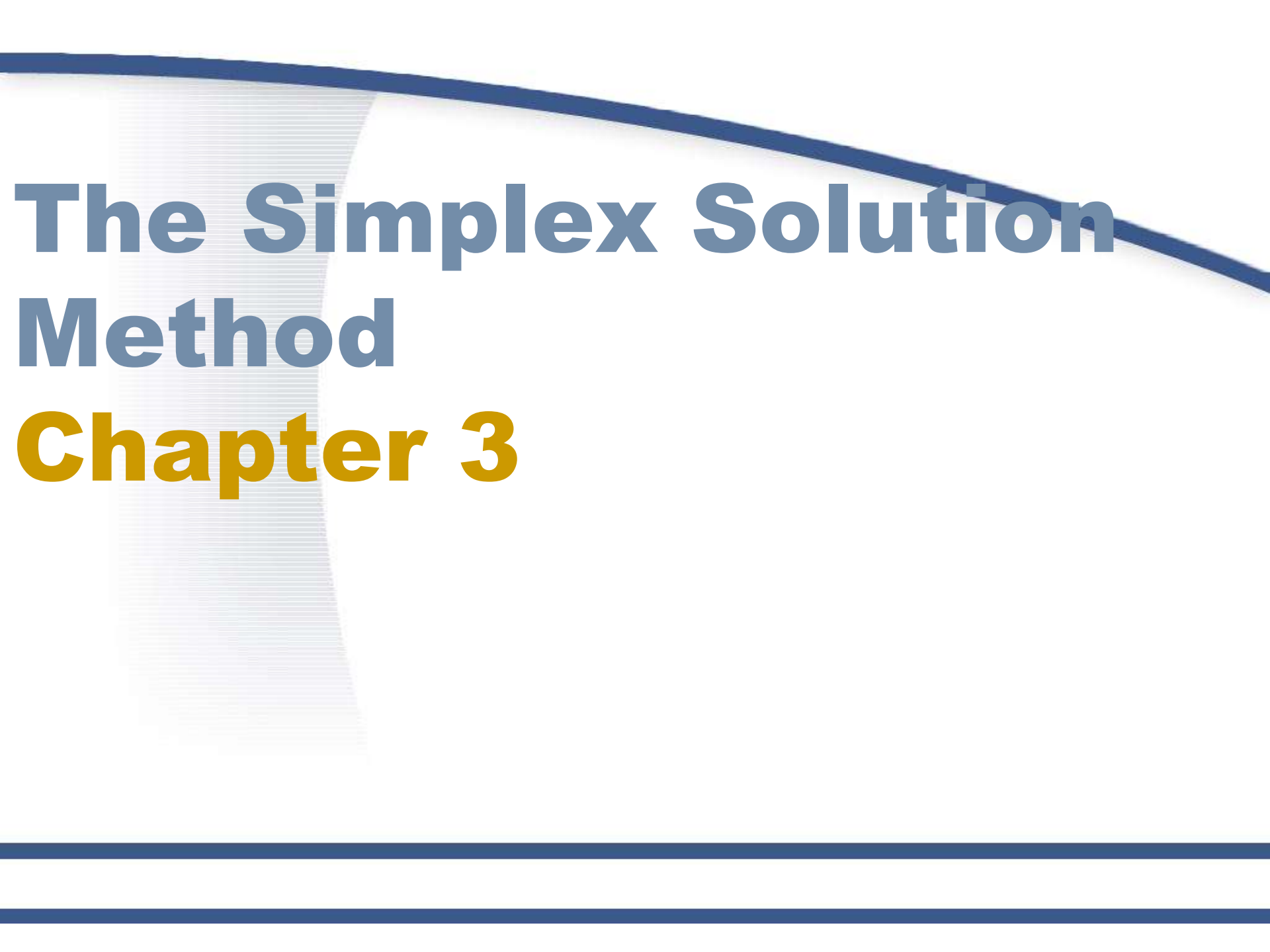




The Simplex Solution Method

Chapter 3

Linear Programming Model

maximize $Z = \$40x_1 + 50x_2$

subject to

$$x_1 + 2x_2 \leq 40 \text{ (labor, hr)}$$

$$4x_1 + 3x_2 \leq 120 \text{ (clay, lb)}$$

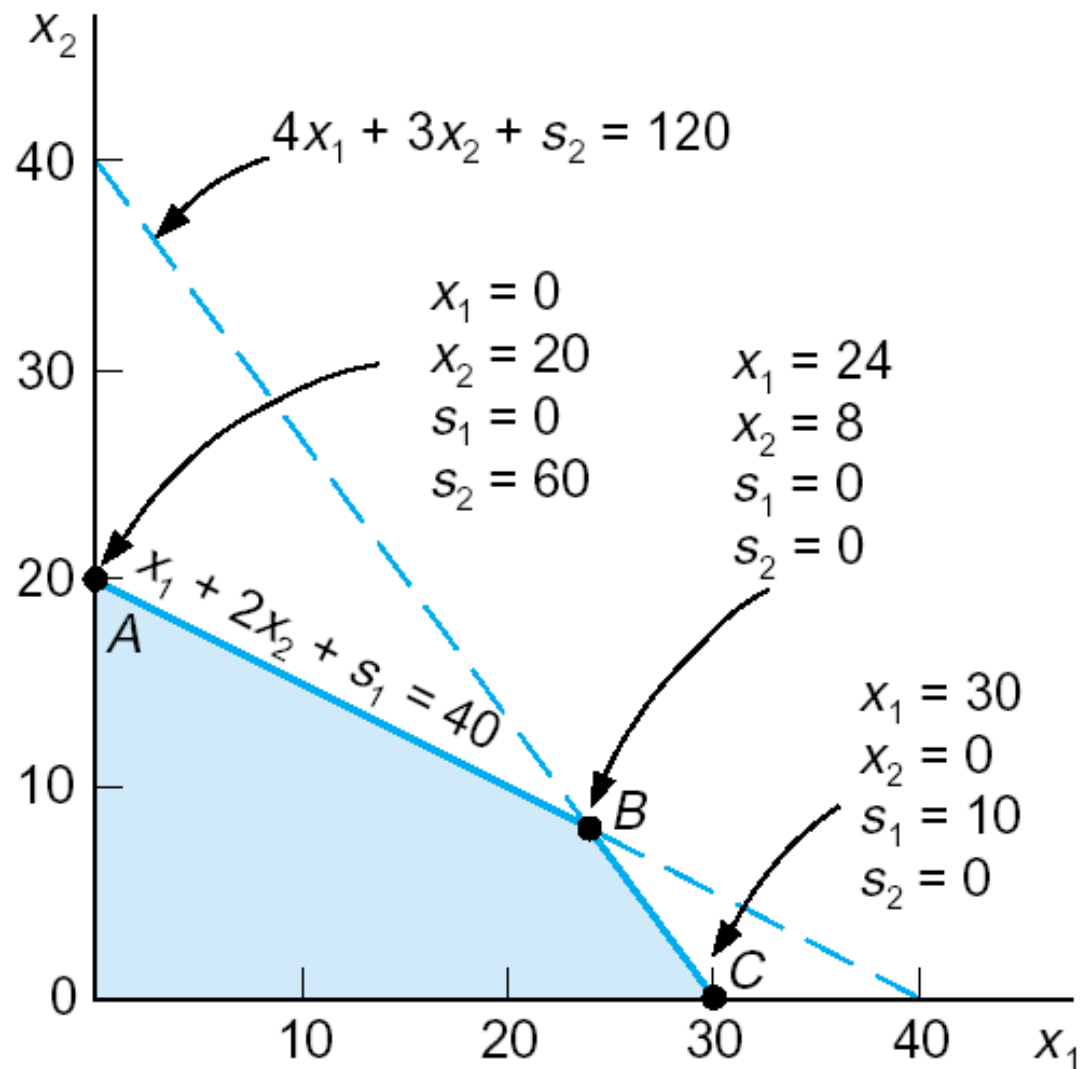
$$x_1, x_2 \geq 0$$

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$
subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$



c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
	z_j					
	$c_j - z_j$					

c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$
 subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$
 subject to

$$\begin{aligned}
 x_1 + 2x_2 + s_1 &= 40 \\
 4x_1 + 3x_2 + s_2 &= 120 \\
 x_1, x_2, s_1, s_2 &\geq 0
 \end{aligned}$$

c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$

subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$
 subject to

$$\begin{aligned} x_1 + 2x_2 + s_1 &= 40 \\ 4x_1 + 3x_2 + s_2 &= 120 \end{aligned}$$

$$x_1, x_2, s_1, s_2 \geq 0$$

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$

subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$

subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$

subject to

$$x_1 + 2x_2 + s_1 = 40$$

$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	Z_j	0	0	0	0	0
	$c_j - Z_j$		40	50	0	0

maximize $Z = 40x_1 + 50x_2 + 0s_1 + 0s_2$

subject to

$$x_1 + 2x_2 + s_1 = 40$$

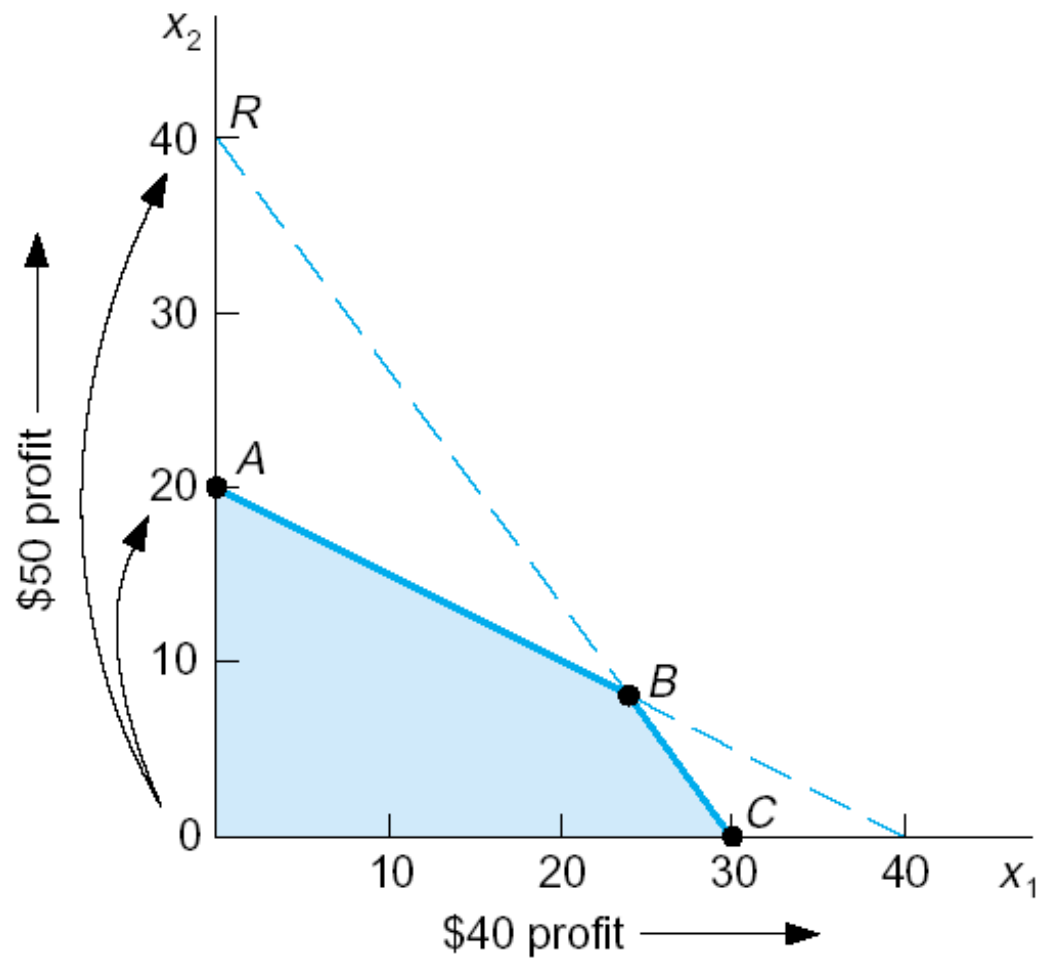
$$4x_1 + 3x_2 + s_2 = 120$$

$$x_1, x_2, s_1, s_2 \geq 0$$

**Pivot
Column**

c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j - z_j$		40	50	0	0

**Largest
 $c_j - z_j$ value**



			Pivot Column			
	Basic Variables	Quantity	40	50	0	0
c_j			x_1	x_2	s_1	s_2
0	s_1	40	1	2	1	0
0	s_2	120	4	3	0	1
	z_j	0	0	0	0	0
	$c_j - z_j$		40	50	0	0

$40 / 2 = 20$

$120 / 3 = 40$

Largest $c_j - z_j$ value

		Pivot Row		Pivot Column			
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	
0	s_1	40	1	2	1	0	
0	s_2	120	4	3	0	1	
	z_j	0	0	0	0	0	
	$c_j - z_j$		40	50	0	0	

Smallest value

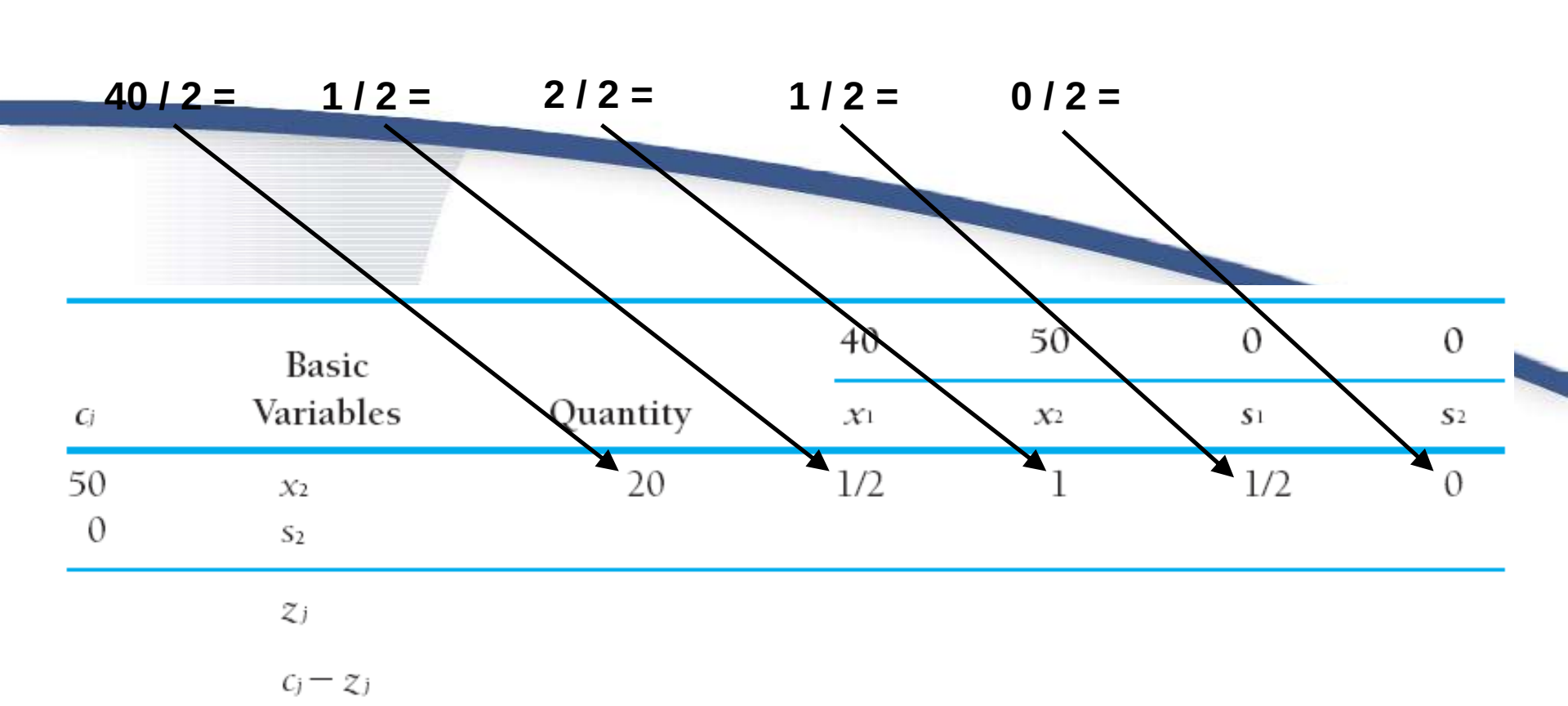
Pivot Number

Largest $c_j - z_j$ value

		Entering Basic Variable			
c_j	Basic Variables	40	50	0	0
			x_2	s_1	s_2
0	s_1	40	1	1	0
	s_2	120	4	0	1
	z_j	0	0	0	0
	$c_j - z_j$	40	50	0	0

← Leaving
Basic
Variable

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
50	x_2					
0	s_2					
	z_j					
	$c_j - z_j$					



C_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
50	x_2	20	$1/2$	1	$1/2$	0
0	s_2					
	Z_j					
	$C_j - Z_j$					

$$\text{New tableau pivot row values} = \frac{\text{Old tableau pivot row values}}{\text{Pivot number}}$$

$$120 - (3 \times 20) =$$

$$3 - (3 \times 1) = 0$$

$$1 - (3 \times 0) =$$

$$4 - (3 \times 1/2) =$$

$$0 - (3 \times 1/2) =$$

		40	50	0	0
C_j	Basic Variables	x_1	x_2	s_1	s_2
50	x_2	1/2	1	1/2	0
0	s_2	5/2	0	-3/2	1
Z_j					
$C_j - Z_j$					

New tableau row values =

Old tableau row values - $\left(\begin{array}{l} \text{Corresponding} \\ \text{coefficients in X} \\ \text{pivot column} \end{array} \right) \left(\begin{array}{l} \text{Corresponding} \\ \text{new tableau} \\ \text{pivot row value} \end{array} \right)$

			40	50	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
50	x_2	20	$1/2$	1	$1/2$	0
0	s_2	60	$5/2$	0	$-3/2$	1
z_j						
$c_j - z_j$						

C_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
50	x_2	20	$1/2$	1	$1/2$	0
0	s_2	60	$5/2$	0	$-3/2$	1
	Z_j	1,000	25	50	25	0
	$C_j - Z_j$		15	0	-25	0

C_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
50	x_2	20	1/2	1	1/2	0
0	s_2	60	5/2	0	-3/2	1
	Z_j	1,000	25	50	25	0
	$C_j - Z_j$		15	0	-25	0

C_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
50	x_2	20	1/2	1	1/2	0
0	s_2	60	5/2	0	-3/2	1
	Z_j	1,000	25	50	25	0
	$C_j - Z_j$		15	0	-25	0

			Pivot Column			
Pivot Row						
C_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
50	x_2	20	$1/2$	1	$1/2$	0
0	s_2	60	$5/2$	0	$-3/2$	1
	Z_j	1,000	25	50	25	0
	$C_j - Z_j$		15	0	-25	0

Smallest value

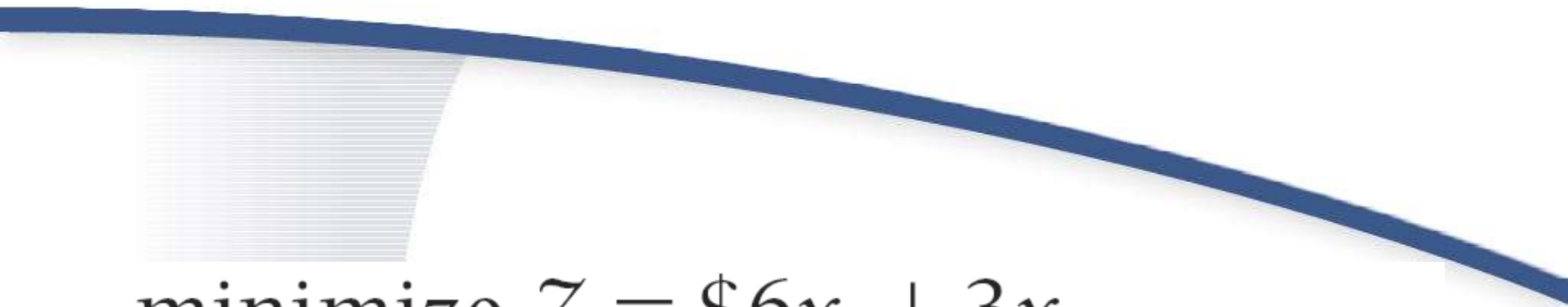
Pivot Number

Largest $C_j - Z_j$ value



c_j	Basic Variables	Quantity	40	50	0	0
			x_1	x_2	s_1	s_2
50	x_2	8	0	1	$4/5$	$-1/5$
40	x_1	24	1	0	$-3/5$	$2/5$
	z_j	1,360	40	50	16	6
	$c_j - z_j$		0	0	-16	-6

Optimal solution



minimize $Z = \$6x_1 + 3x_2$

subject to

$2x_1 + 4x_2 \geq 16$ lb of nitrogen

$4x_1 + 3x_2 \geq 24$ lb of phosphate



minimize $Z = 6x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$
subject to

$$2x_1 + 4x_1 - s_1 + A_1 = 16$$

$$4x_1 + 3x_2 - s_2 + A_2 = 24$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

			6	3	0	0	M	M
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	A_1	A_2
M	A_1	16	2	4	-1	0	1	0
M	A_2	24	4	3	0	-1	0	1
	z_j	40M	6M	7M	-M	-M	M	M
	$z_j - c_j$		6M - 6	7M - 3	-M	-M	0	0

minimize $Z = 6x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$

subject to

$$2x_1 + 4x_2 - s_1 + A_1 = 16$$

$$4x_1 + 3x_2 - s_2 + A_2 = 24$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

c_j	Basic Variables	Quantity	6	3	0	0	M	M
			x_1	x_2	s_1	s_2	A_1	A_2
M	A_1	16	2	4	-1	0	1	0
M	A_2	24	4	3	0	-1	0	1
	z_j	40M	6M	7M	-M	-M	M	M
	$z_j - c_j$		6M - 6	7M - 3	-M	-M	0	0

minimize $Z = 6x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$

subject to

$$2x_1 + 4x_2 - s_1 + A_1 = 16$$

$$4x_1 + 3x_2 - s_2 + A_2 = 24$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

c_j	Basic Variables	Quantity	6	3	0	0	M	M
			x_1	x_2	s_1	s_2	A_1	A_2
M	A_1	16	2	4	-1	0	1	0
M	A_2	24	4	3	0	-1	0	1
	z_j	40M	6M	7M	-M	-M	M	M
	$z_j - c_j$		6M - 6	7M - 3	-M	-M	0	0

$$24 - (3 \times 4) = 12$$

c_j	Basic Variables	Quantity	6	3	0	0	M
			x_1	x_2	s_1	s_2	A_2
3	x_2	4	1/2	1	-1/4	0	0
M	A_2	12	5/2	0	3/4	-1	1
	z_j	12M + 12	5M/2 + 3/2	3	-3/4 + 3M/4	-M	M
	$z_j - c_j$		5M/2 - 9/2	0	-3/4 + 3M/4	-M	0

c_j	Basic Variables	Quantity	6	3	0	0	M
			x_1	x_2	s_1	s_2	A_2
3	x_2	4	1/2	1	-1/4	0	0
M	A_2	12	5/2	0	3/4	-1	1
	z_j	$12M + 12$	$5M/2 + 3/2$	3	$-3/4 + 3M/4$	$-M$	M
	$z_j - c_j$		$5M/2 - 9/2$	0	$-3/4 + 3M/4$	$-M$	0

c_j	Basic Variables	Quantity	6	3	0	0
			x_1	x_2	s_1	s_2
3	x_2	$8/5$	0	1	$-2/5$	$1/5$
6	x_1	$24/5$	1	0	$3/10$	$-2/5$
	z_j	$168/5$	6	3	$3/5$	$-9/5$
	$z_j - c_j$		0	0	$3/5$	$-9/5$

			6	3	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2
3	x_2	8	$4/3$	1	0	$-1/3$
0	s_1	16	$10/3$	0	1	$-4/3$
	z_j	24	4	3	0	-1
	$z_j - c_j$		-2	0	0	-1

Optimal solution



To summarize, the adjustments necessary to apply the simplex method to a minimization problem are as follows:

1. Transform all $\geq$ constraints to equations by subtracting a surplus variable and adding an artificial variable.
 2. Assign a c_j value of M to each artificial variable in the objective function.
 3. Change the $c_j - z_j$ row to $z_j - c_j$.
- 

A leather shop makes custom-designed, hand-tooled briefcases and luggage. The shop makes a \$400 profit from each **briefcase** and a \$200 profit from piece of **luggage**.

The shop has a contract to provide a store with exactly 30 items per month.

A tannery supplies the shop with at least 80 square yards of leather per month.

The shop must use at least this amount but can order more.

Each **briefcase** requires 2 square yards of leather, each piece of **luggage** requires 8 square yards of leather.



From past performance, the shop owners know they cannot make more than 20 briefcases per month.

They want to know the number of briefcases and pieces of luggage to produce in order to **maximize profit**.

Example – Summary of Simplex Method

maximize $Z = \$400x_1 + 200x_2$

subject to

$x_1 + x_2 = 30$ contracted items

$2x_1 + 8x_2 \geq 80$ yd² of leather

$x_1 \leq 20$ briefcases

$x_1, x_2 \geq 0$

1. Transform the model constraint inequalities into equations.

Constraint	Adjustment	Objective Function Coefficient	
		Maximization	Minimization
$\leq$	Add a slack variable	0	0
$=$	Add an artificial variable	$-M$	M
$\geq$	Subtract a surplus variable	0	0
	and add an artificial variable	$-M$	M

maximize $Z = 400x_1 + 200x_2 + 0s_1 + 0s_2 - MA_1 - MA_2$
subject to

$$x_1 + x_2 + A_1 = 30$$

$$2x_1 + 8x_2 - s_1 + A_2 = 80$$

$$x_1 + s_2 = 20$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

2. Set up initial tableau for the basic feasible solution at the origin and compute the z_j and $c_j - z_j$ row values.

			400	200	0	0	$-M$	$-M$
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	A_1	A_2
$-M$	A_1	30	1	1	0	0	1	0
$-M$	A_2	80	2	8	-1	0	0	1
0	s_2	20	1	0	0	1	0	0
	z_j	$-110M$	$-3M$	$-9M$	M	0	$-M$	$-M$
	$c_j - z_j$		$400 + 3M$	$200 + 9M$	$-M$	0	0	0

3. Determine the pivot column by selecting the column with the highest positive value in the $c_j - z_j$ row.

			400	200	0	0	$-M$	$-M$
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	A_1	A_2
$-M$	A_1	30	1	1	0	0	1	0
$-M$	A_2	80	2	8	-1	0	0	1
0	s_2	20	1	0	0	1	0	0
	z_j	$-110M$	$-3M$	$-9M$	M	0	$-M$	$-M$
	$c_j - z_j$		$400 + 3M$	$200 + 9M$	$-M$	0	0	0

4. Determine the pivot row by dividing the quantity column values by the pivot column values and selecting the row with the minimum nonnegative quotient.

C_j	Basic Variables	Quantity	400	200	0	0	$-M$	$-M$
			x_1	x_2	s_1	s_2	A_1	A_2
$-M$	A_1	30	1	1	0	0	1	0
$-M$	A_2	80	2	8	-1	0	0	1
0	s_2	20	1	0	0	1	0	0
z_j		$-110M$	$-3M$	$-9M$	M	0	$-M$	$-M$
$C_j - z_j$			$400 + 3M$	$200 + 9M$	$-M$	0	0	0

5. Compute the pivot row values using the formula:

new tableau pivot row values = $\frac{\text{old tableau pivot row values}}{\text{pivot number}}$

	Basic Variables	Quantity	400	200	0	0	-M	-M
C_j			x_1	x_2	s_1	s_2	A_1	A_2
-M	A_1	30	1	1	0	0	1	0
-M	A_2	80	2	8	-1	0	0	1
0	s_2	20	1	0	0	1	0	0
z_j		-110M	-3M	-9M	M	0	-M	-M
$C_j - z_j$			400 + 3M	200 + 9M	-M	0	0	0

$80 / 8 = 10$

$2 / 8 = 1/4$

$8 / 8 = 1$

$-1 / 8 = -1/8$

$1 / 8 = 1/8$

6. Compute all other row values using the formula:

$$\text{new tableau row values} = \text{old tableau row values} - \left(\begin{array}{c} \text{corresponding} \\ \text{coefficients in} \\ \text{pivot column} \end{array} \times \begin{array}{c} \text{corresponding} \\ \text{new tableau} \\ \text{pivot row value} \end{array} \right)$$

			400	200	0	0	-M
C_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	A_1
-M	A_1	20	3/4	0	1/8	0	1
200	x_2	20	1/4	1	-1/8	0	0
0	s_2	20	1	0	0	1	0
Z_j		2,000 - 20M	50 - 3M/4	200	-25 - M/8	0	-M
$C_j - Z_j$			350 + 31M/4	0	25 + M/8	0	0

$$30 - (1 \times 10) = 20$$

$$20 - (0 \times 10) = 20$$

$$1 - (0 \times 0) = 1$$

7. Compute the new z_j and $c_j - z_j$ rows.

c_j	Basic Variables	Quantity	400 x_1	200 x_2	0 s_1	0 s_2	$-M$ A_1
$-M$	A_1	20	$3/4$	0	$1/8$	0	1
200	x_2	10	$1/4$	1	$-1/8$	0	0
0	s_2	20	1	0	0	1	0
z_j		$2,000 - 20M$	$50 - 3M/4$	200	$-25 - M/8$	0	$-M$
$c_j - z_j$			$350 + 3M/4$	0	$25 + M/8$	0	0

8. Determine whether or not the new solution is optimal by checking the $c_j - z_j$ row. If all $c_j - z_j$ row values are zero or negative, the solution is optimal. If a positive value exists, return to step 3 and repeat the simplex steps.

			400	200	0	0	-M
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	A_1
-M	A_1	20	3/4	0	1/8	0	1
200	x_2	10	1/4	1	-1/8	0	0
0	s_2	20	1	0	0	1	0
	z_j	2,000 - 20M	50 - 3M/4	200	-25 - M/8	0	-M
	$c_j - z_j$		350 + 3M/4	0	25 + M/8	0	0

The Third Simplex Tableau

C_j	Basic Variables	Quantity	400	200	0	0	$-M$
			x_1	x_2	s_1	s_2	A_1
$-M$	A_1	5	0	0	$1/8$	$-3/4$	1
200	x_2	5	0	1	$-1/8$	$-1/4$	0
400	x_1	20	1	0	0	1	0
z_j			400	200	$-25 - M/8$	$350 + 3M/4$	$-M$
$C_j - z_j$			0	0	$25 + M/8$	$-350 - 3M/4$	0

The Optimal Simplex Tableau

C_j	Basic Variables	Quantity	400	200	0	0
			x_1	x_2	s_1	s_2
0	s_1	40	0	0	1	-6
200	x_2	10	0	1	0	-1
400	x_1	20	1	0	0	1
	z_j	10,000	400	200	0	200
	$C_j - z_j$		0	0	0	-200

$x_1 = 20$ briefcases

$x_2 = 10$ pieces of luggage

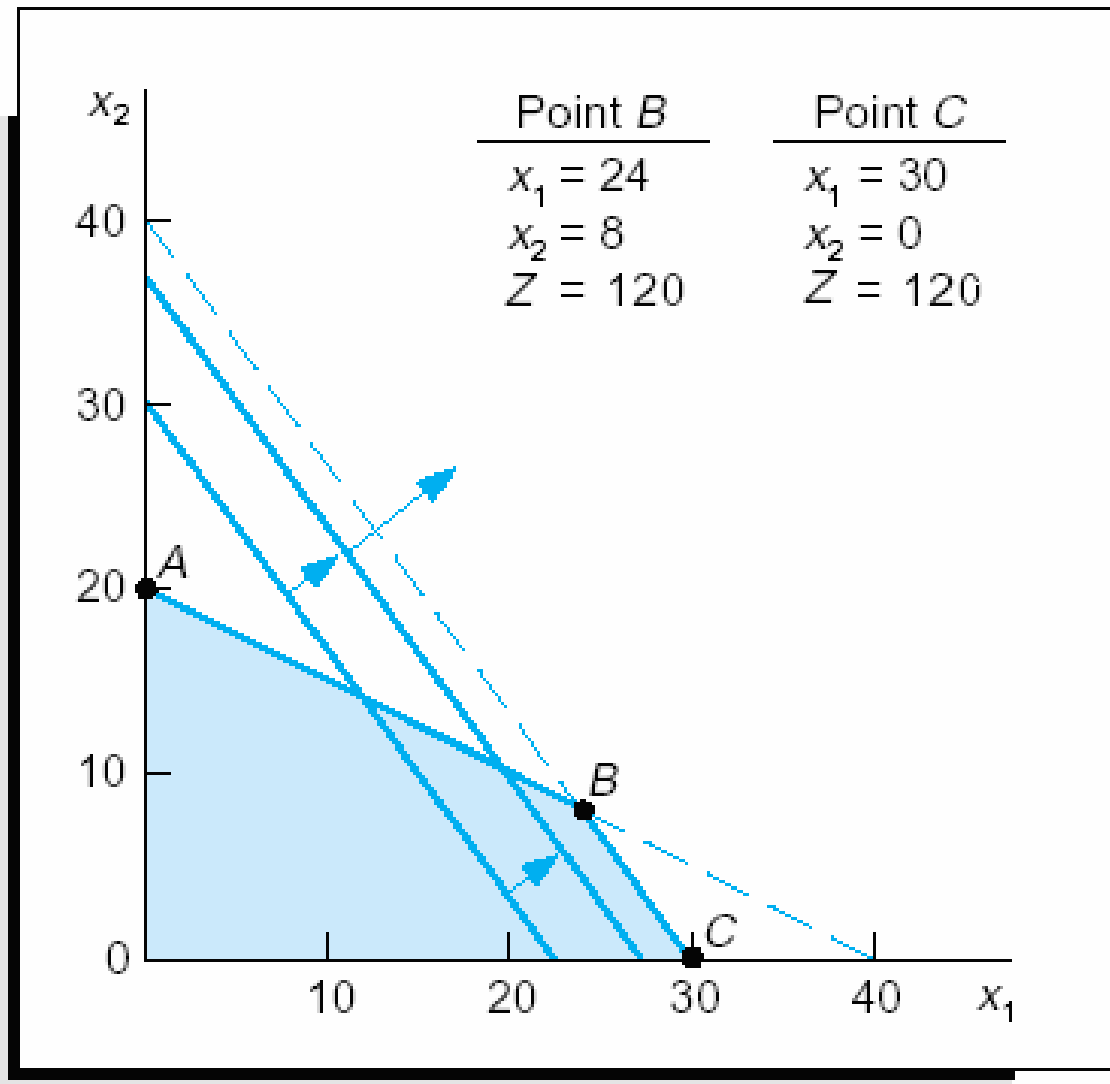
$s_1 = 40$ extra square yards of leather

$Z = \$10,000$ profit per month



Irregular Types of Linear Programming Problems





The Multiple Optimal Solutions

c_j	Basic Variables	Quantity	40	30	0	0
			x_1	x_2	s_1	s_2
0	s_1	10	0	$5/4$	1	$-1/4$
40	x_1	30	1	$3/4$	0	$1/4$
	z_j	1,200	40	30	0	10
	$c_j - z_j$		0	0	0	-10

Maximize $Z = 5x_1 + 3x_2$

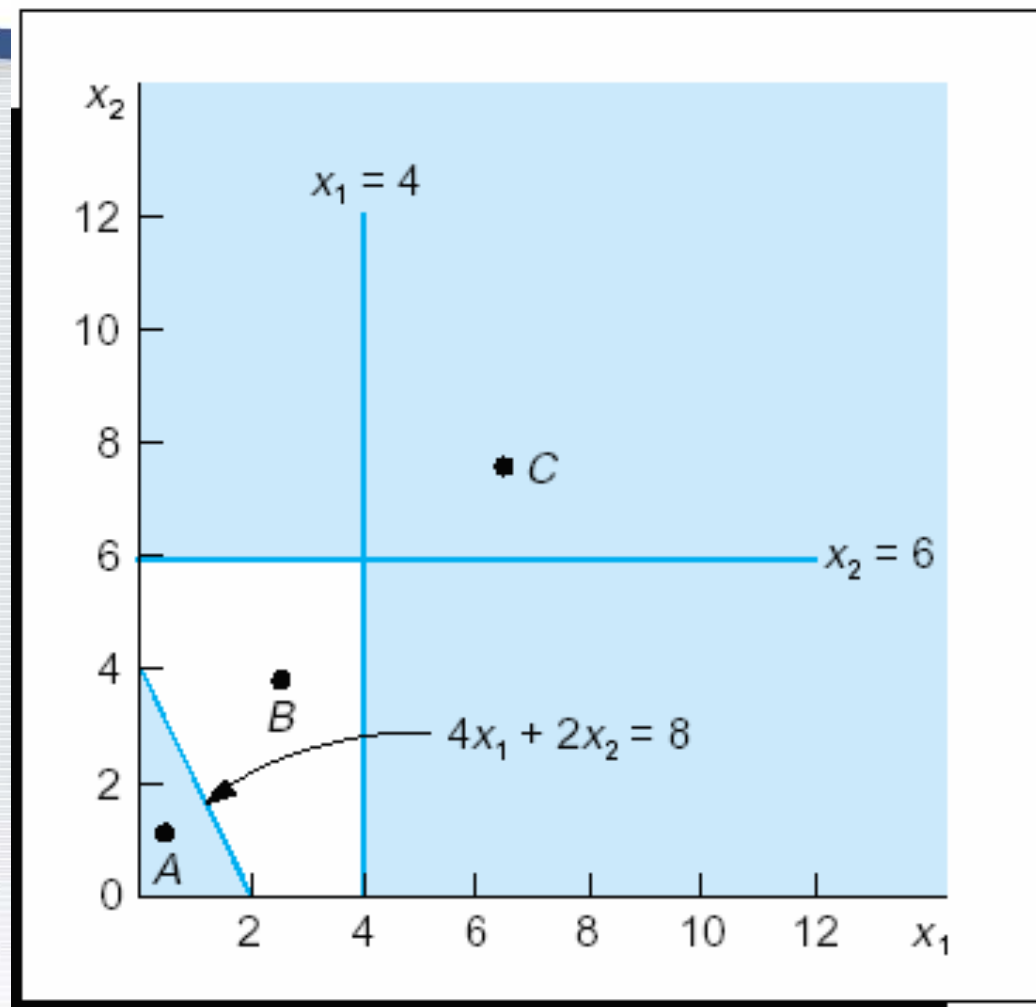
Subject to:

$$4x_1 + 2x_2 \leq 8$$

$$x_1 \geq 4$$

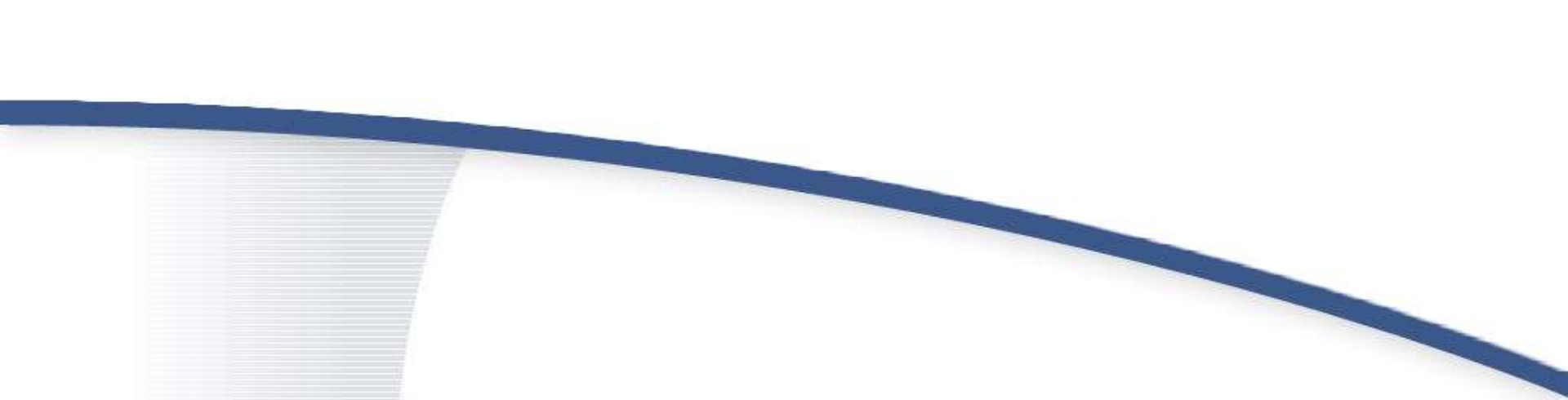
$$x_2 \geq 6$$

$$x_1, x_2 \geq 0$$



An Infeasible Problem

c_j	Basic Variables	Quantity	5	3	0	0	0	-M	-M
			x_1	x_2	s_1	s_2	s_3	A_1	A_2
3	x_1	4	2	1	1/2	0	0	0	0
-M	A_1	4	1	0	0	-1	0	1	0
-M	A_2	2	-2	0	-1/2	0	-1	0	1
	z_j	$12 - 6M$	$6 + M$	3	$3/2 + M/2$	M	M	-M	-M
	$c_j - z_j$		$-1 - M$	0	$-3/2 - M/2$	-M	-M	0	0

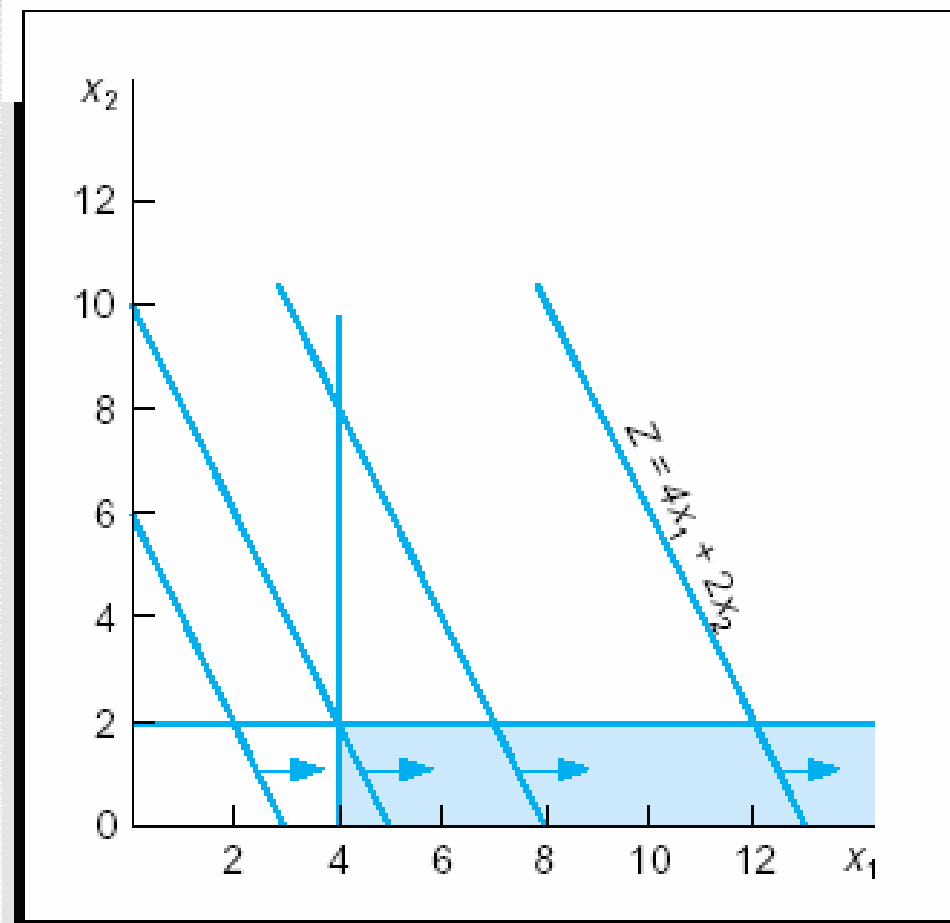


maximize $Z = 4x_1 + 2x_2$
subject to

$$x_1 \geq 4$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



An Unbounded Problem

C_j	Basic Variables	Quantity	4	2	0	0	
			x_1	x_2	s_1	s_2	
4	x_1	4	1	0	-1	0	$4 \div -1 = -4$
0	s_2	2	0	1	0	1	$2 \div 0 = \infty$
	Z_j	16	4	0	-4	0	
	$C_j - Z_j$		0	2	4	0	

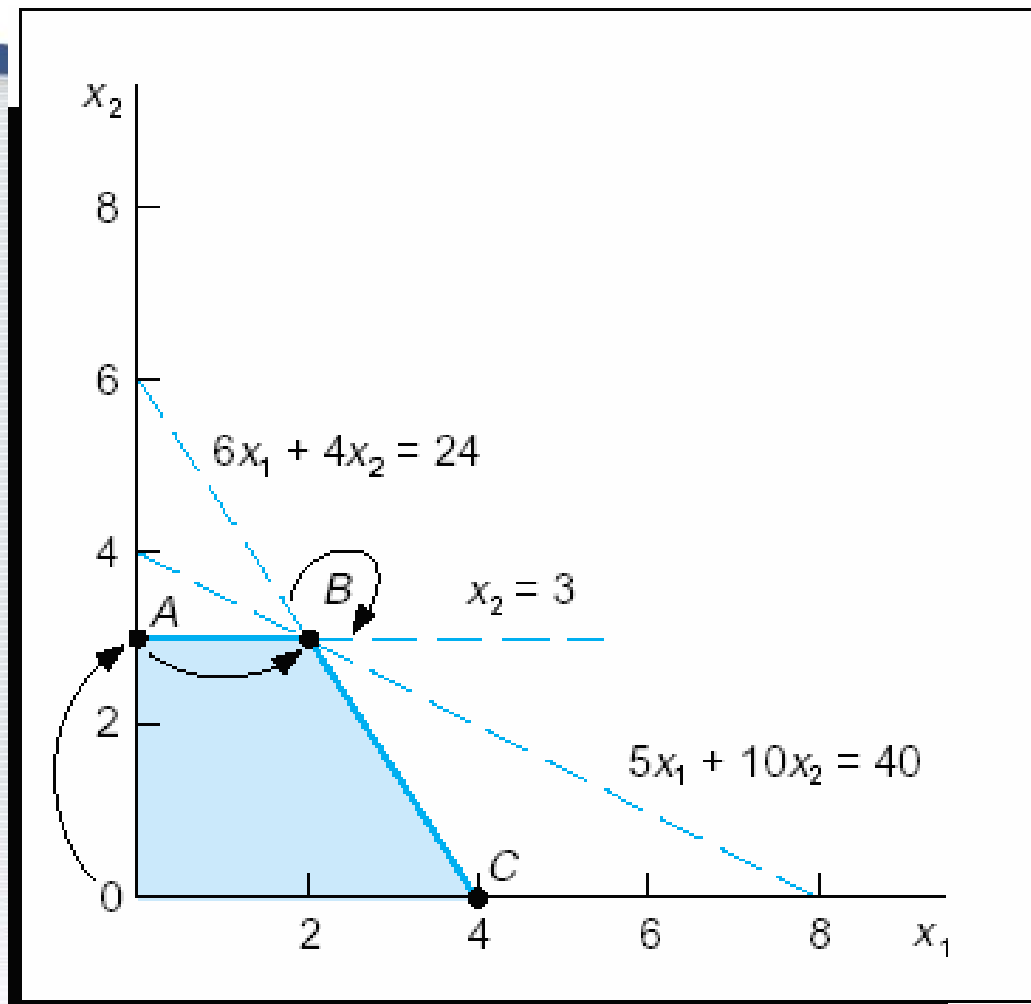
A pivot row cannot be selected for an unbounded problem

One of the two tied columns should be selected arbitrarily.

Tie for the Pivot Row Degeneracy

C_j	Basic Variables	Quantity	4	6	0	0	0	
			x_1	x_2	s_1	s_2	s_3	
0	s_1	12	6	0	1	-4	0	$12 \div 6 = 2$
6	x_2	3	0	1	0	1	0	} Tie
0	s_3	10	5	0	0	-10	1	$10 \div 5 = 2$
	Z_j	18	0	6	0	6	0	
	$C_j - Z_j$		4	0	0	-6	0	

			4	6	0	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	s_3
0	s_1	0	0	0	1	8	$-6/5$
6	x_2	3	0	1	0	1	0
4	x_1	2	1	0	0	-2	$1/5$
	z_j	26	4	6	0	-2	$4/5$
	$c_j - z_j$		0	0	0	2	$-4/5$

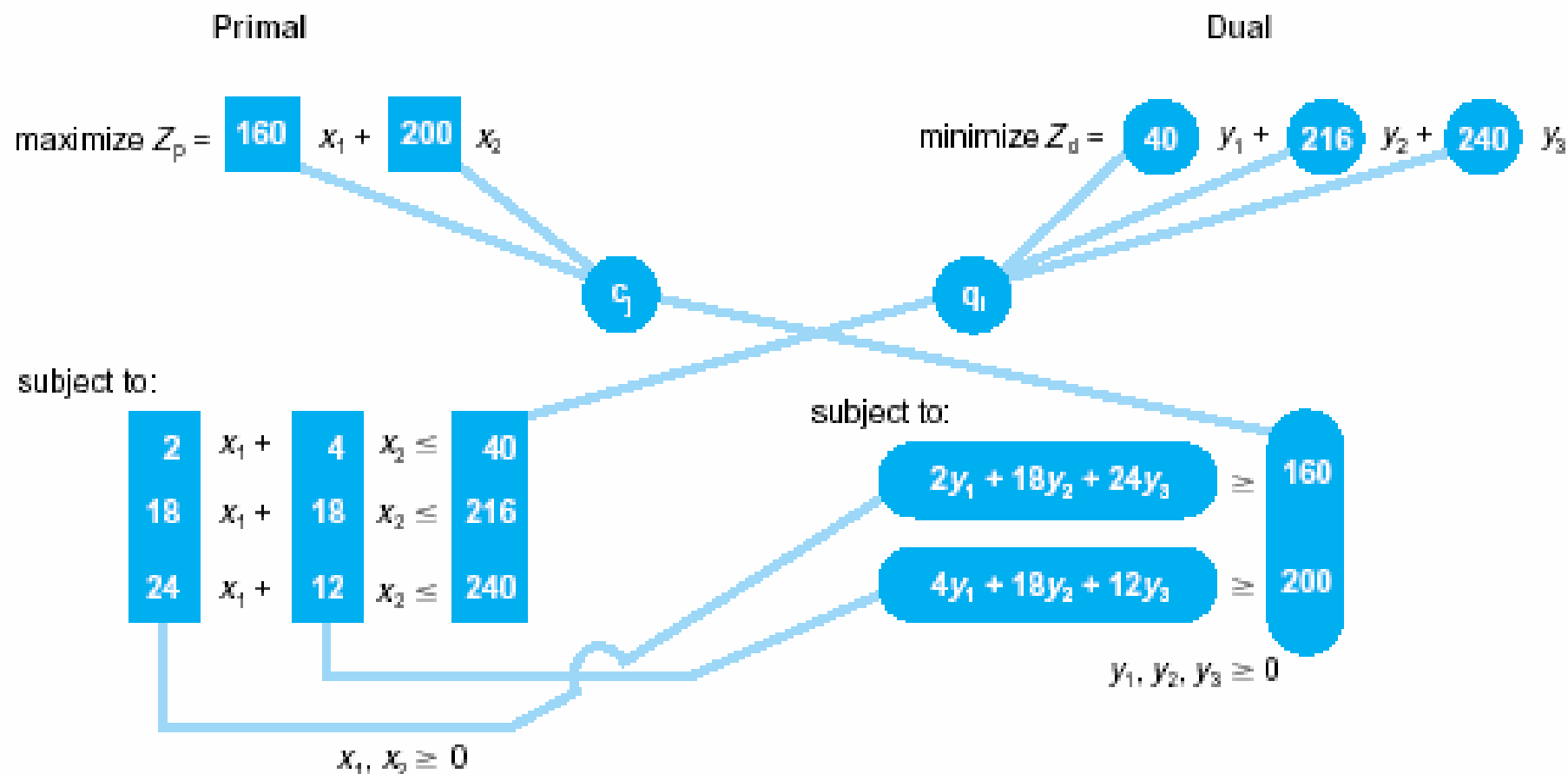


Standard form for simplex solution requires positive right-hand-side values.

$$- 6x_1 + 2x_2 \geq - 30$$

$$(-1) (- 6x_1 + 2x_2 \geq -30)$$

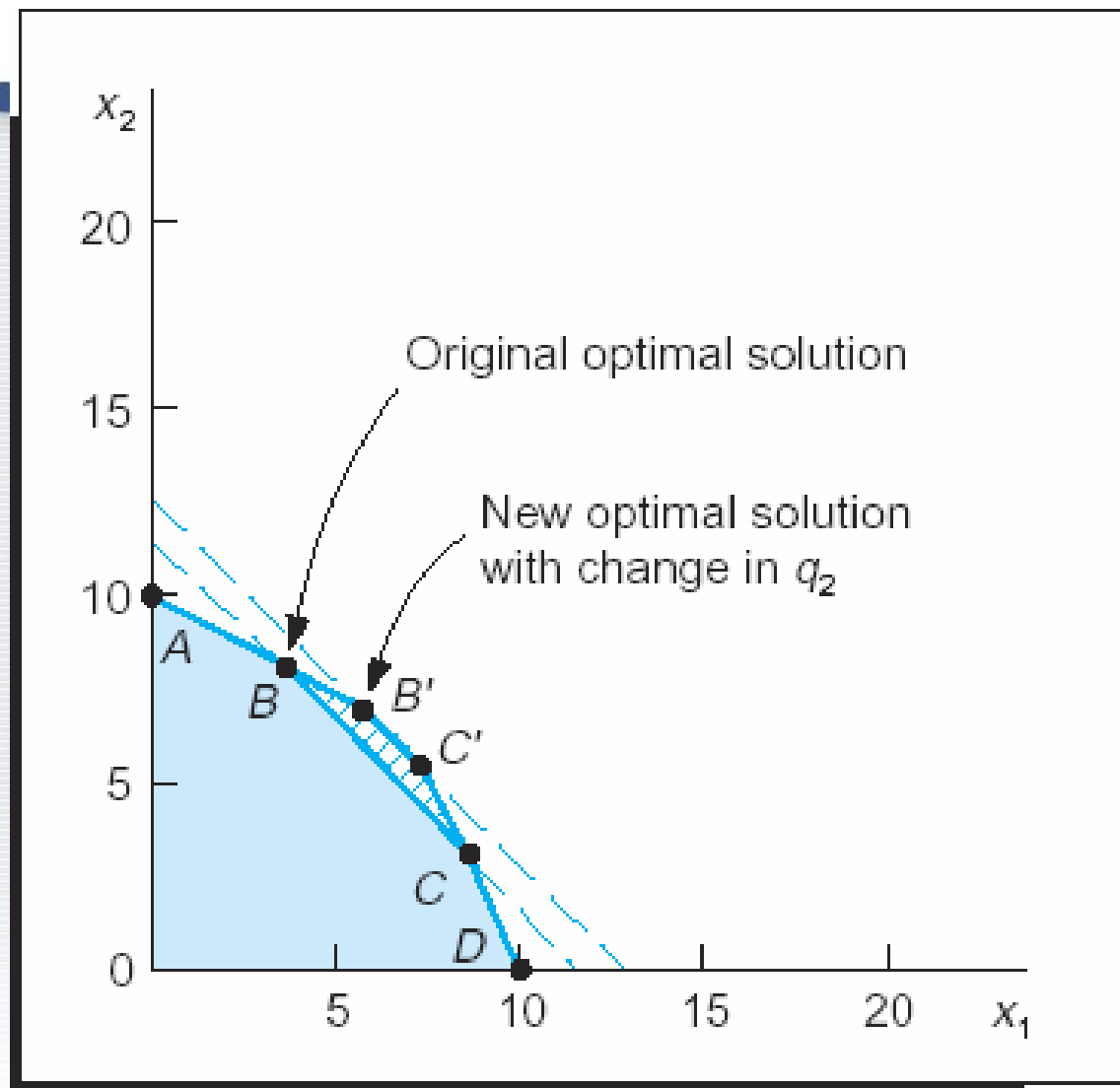
$$6x_1 - 2x_2 \leq 30$$



C_j	Basic Variables	Quantity	160	200	0	0	0
			x_1	x_2	s_1	s_2	s_3
200	x_2	8	0	1	$1/2$	$-1/18$	0
160	x_1	4	1	0	$-1/2$	$1/9$	0
0	s_3	48	0	0	6	-2	1
	z_j	2,240	160	200	20	$20/3$	0
	$C_j - z_j$		0	0	-20	$-20/3$	0

c_j	Basic Variables	Quantity	$160 + \Delta$	200	0	0	0
			x_1	x_2	s_1	s_2	s_3
200	x_2	8	0	1	$1/2$	$-1/18$	0
$160 + \Delta$	x_1	4	1	0	$-1/2$	$1/9$	0
0	s_3	48	0	0	6	-2	1
z_j			$160 + \Delta$	200	$20 - \Delta/2$	$20/3 + \Delta/9$	0
$c_j - z_j$			0	0	$-20 + \Delta/2$	$-20/3 - \Delta/9$	0

			160	$200 + \Delta$	0	0	0
c_j	Basic Variables	Quantity	x_1	x_2	s_1	s_2	s_3
$200 + \Delta$	x_2	8	0	1	$1/2$	$-1/18$	0
160	x_1	4	1	0	$-1/2$	$1/9$	0
0	s_3	48	0	0	6	-2	1
	z_j	$2,240 + 8\Delta$	160	$200 + \Delta$	$20 + \Delta/2$	$20/3 - \Delta/18$	0
	$c_j - z_j$		0	0	$-20 - \Delta/2$	$-20/3 + \Delta/18$	0



C_j	Basic Variables	Quantity	160	200	0	0	0
			x_1	x_2	s_1	s_2	s_3
0	s_1	$40 + 1\Delta$	2	4	1	0	0
0	s_2	$216 + 0\Delta$	18	18	0	1	0
0	s_3	$240 + 0\Delta$	24	12	0	0	1
	z_j	0	0	0	0	0	0
	$C_j - z_j$		160	200	0	0	0

c_j	Basic Variables	Quantity	160	200	0	0	0
			x_1	x_2	s_1	s_2	s_3
200	x_2	$8 + \Delta/2$	0	1	$1/2$	$-1/18$	0
160	x_1	$4 - \Delta/2$	1	0	$-1/2$	$1/9$	0
0	s_3	$48 + 6\Delta$	0	0	6	-2	1
z_j			160	200	20	$20/3$	0
$c_j - z_j$			0	0	-20	$-20/3$	0